

CSCI 246 – Discrete Structures

Mike Wojnowicz

Fall 2026

Montana State University

Course Overview

Why Active Learning?

About Me

Discrete Math In Practice: Examples
From Machine Learning

Virtues Of Discrete Math

How To Read Math

What is Discrete Math?

dis·crete / dis'krēt.

Adjective: Individually separate and distinct.

Synonyms: separate - detached - distinct - abstract.

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See [Canvas](#) for syllabus.

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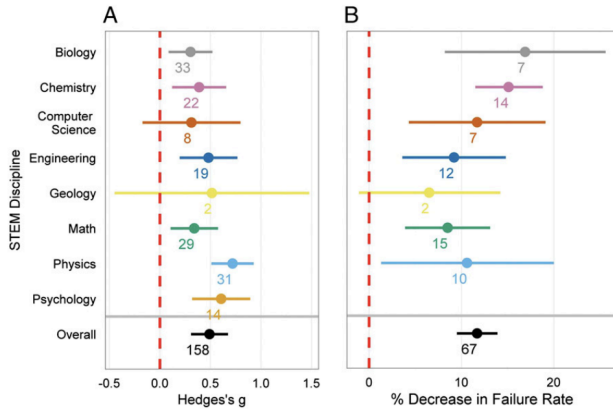
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Freeman et al. (2014). *Proceedings of the National Academy of Sciences*.

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Michael Wojnowicz

Assistant Professor, GSoC @ MSU

Research Associate, Biostats @ Harvard

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Distinguished Data Scientist @ Cylance

(cybersecurity start-up, sold for \$1.4b)

PhD @ Cornell

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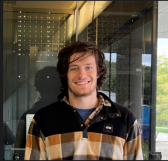
Research interests

- Probabilistic machine learning
- Scalable inference
- Modeling spatiotemporal data

MSU Probabilistic Machine Learning (ProbML) Lab

Lab Members

Jack Ruder (Ph.D. Student)



Diffusion models for time series

Kunal Das (Postdoc)



Spatiotemporal Causal Inference

Landry Clarke (Undergrad)



Deep Hierarchical Gaussian N-Grams

* Winner: MSU Undergraduate Scholars research award

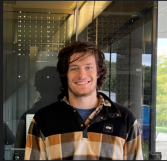
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My role: 30% teaching, 60% research, 10% service.

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Problem #1: Continuous Authentication



Continuous authentication:

continuously verify a user's identity
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Difficulty: Many 1000s of computer processes, sparsely observed.

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	Caffeine
	Activity Monitor
	Google Chrome Helper (GPU)
	Texifier
	Microsoft Outlook

Subgoal: Learn computer process start sequences which characterize the typical behavior of a user.

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Modeling Desiderata: probabilistic model with fast, reliable training

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The solution to this problem used almost every content area we're covering in this course

Discrete probability and counting

Complexity (Big-O notation)

Recursions

Graph theory

Proof techniques

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Can you explain why?

Problem #2: Early Cancer Detection

Healthy cell

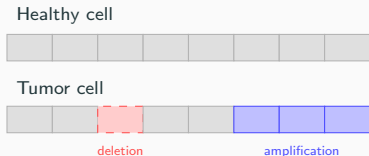


Tumor cell

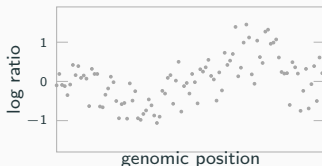


Somatic copy number alterations (SCNAs): tumors gain and lose whole *segments* of DNA, carving the genome into contiguous blocks, each at its own copy number.

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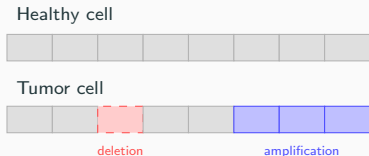


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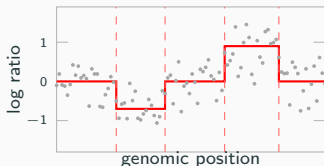
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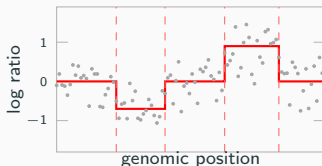
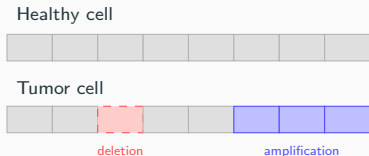
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Modeling Desiderata: probabilistic model that is accurate at low signal-to-noise, yet fast enough to scale

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Tips for Reading Math Textbooks

Work through examples.

- Don't just skim (or worse, skip!) the examples. Instead, devote more of your focus to the examples. Working through them and verbalizing the main ideas behind them is a great way to test to make sure you're understanding what you're reading.
- Read with a pencil and paper in your hand and when you get to an example, work it out! Try not to look at the solution until you are done.
- When checking your work, make sure you understand each step and why.
- Practice the examples BEFORE you attempt homework problems and then try to do your homework without referring back to the example